

CA2 PROJECT – DATA SCIENCE

**ANTI-MONEY LAUNDERING (AML) DETECTION
SYSTEM**

USING MACHINE LEARNING

MSc IT – Part I

Project Report

Submitted By

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Introduction

Money laundering is the process of disguising illegally obtained money so that it appears to originate from legitimate sources. Financial institutions process large volumes of transactions through cash, cards, cheques, electronic transfers and cross-border payments. Identifying potentially suspicious patterns within this volume is an important application of Data Science.

This project develops an Anti-Money Laundering (AML) Detection System using Machine Learning. The project follows a Data Science workflow consisting of data loading, data quality checking, preprocessing, exploratory data analysis, visualization, feature engineering, classification and model evaluation.

The analysis uses transaction records from the Synthetic Anti-Money Laundering Dataset (SAML-D). The target variable, `Is_laundering`, indicates whether a transaction is classified as laundering. A Random Forest Classifier is used to learn patterns from selected transaction characteristics.

Project Overview

The developed system is also connected to a Streamlit application. The application accepts transaction-related inputs and uses the trained model to produce a classification result. The notebook submitted for CA2 documents the underlying Data Science workflow.

Need for the Project

Manual review of a large number of transactions can be time-consuming. A machine learning approach can help organize transaction information, identify patterns and provide an automated first-level classification that can support further investigation.

• Problem Statement

To develop a machine learning-based system that analyzes financial transaction information and classifies transactions as potentially suspicious or non-suspicious using relevant transaction features.

Objectives:-

- To understand and analyze financial transaction data.
- To preprocess the dataset for analysis and machine learning.
- To perform Exploratory Data Analysis (EDA).
- To create meaningful visualizations, including an interactive Plotly visualization.
- To develop a classification model for identifying potentially suspicious transactions.
- To evaluate the model using Accuracy, Precision, Recall, F1-Score and Confusion Matrix.
- To derive data-supported findings and conclusions.

Dataset Description:-

The dataset used in the executed notebook contains 9,984 transaction records and 12 original attributes. After removing the one row with a missing target value, 9,983 records were used for the analysis and model training.

Dataset Item	Observed Value
Analysis records before target cleanup	9,984
Records after target cleanup	9,983
Original attributes	12
Target	Is_laundering
Duplicate rows	0
Missing target values	1

Dataset Features:-

Feature	Description
Time	Time of the transaction.
Date	Date of the transaction.
Sender_account	Identifier of the sending account.
Receiver_account	Identifier of the receiving account.
Amount	Transaction amount.
Payment_currency	Currency used for payment.
Received_currency	Currency received.
Sender_bank_location	Location of the sender's bank.
Receiver_bank_location	Location of the receiver's bank.
Payment_type	Method/type of payment.
Is_laundering	Target variable indicating laundering classification.
Laundering_type	Transaction typology/category.

Data Source:-

The dataset source used for the project is the Synthetic Transaction Monitoring Dataset (SAML-D) available on Kaggle.

Dataset Link: <https://www.kaggle.com/datasets/berkanoztas/synthetic-transaction-monitoring-dataset-aml>

The notebook executed for this submission contains 9,984 records in the local SAML-D.csv used for analysis. The report therefore uses the observed notebook dataset size rather than substituting a different dataset size.

Data Preprocessing:-

The preprocessing stage focused on checking data quality and preparing the variables for analysis and classification.

- The dataset structure, number of rows and number of columns were checked.
- Data types and unique-value counts were examined.
- Duplicate records were checked; no duplicate rows were found.
- Missing values were identified. One missing value was present in several transaction fields, including Receiver_account, Amount, currencies, bank locations, Payment_type, Is_laundering and Laundering_type.
- The single row with a missing target value was removed, resulting in 9,983 records.
- Categorical features were one-hot encoded before Random Forest training.
- Two engineered features were created: Amount_Risk and Payment_Risk.
- Amount_Risk was created from transaction amount ranges, while Payment_Risk was set from payment types classified as cash or cheque.

Feature Engineering:-

Amount_Risk assigns a risk category based on transaction amount: 0 for amounts up to 1,000; 1 for amounts above 1,000 and up to 10,000; 2 for amounts above 10,000 and up to 50,000; and 3 for amounts above 50,000.

Payment_Risk is set to 1 for cash or cheque and 0 otherwise.

Exploratory Data Analysis:-

The executed notebook shows that the dataset is highly imbalanced with respect to the target variable. Out of 9,983 records after target cleanup, 9,970 were classified as non-laundering and 13 as laundering.

Class	Count	Proportion
Non-laundering (0)	9,970	99.8698%
Laundering (1)	13	0.1302%

Transaction Amount Analysis:-

For the 9,983 records with a non-missing target, the transaction amount had a mean of approximately 8,360.91 and a median of approximately 5,821.66. The minimum observed amount was 9.11 and the maximum was 998,453.63. The large difference between the mean and maximum indicates that the amount distribution contains high-value observations.

Payment Type Analysis:-

Payment type contains seven unique categories in the dataset. In the descriptive output, Credit card is the most frequent payment type, appearing 2,064 times among the 9,983 usable target records.

Currency and Location Analysis:-

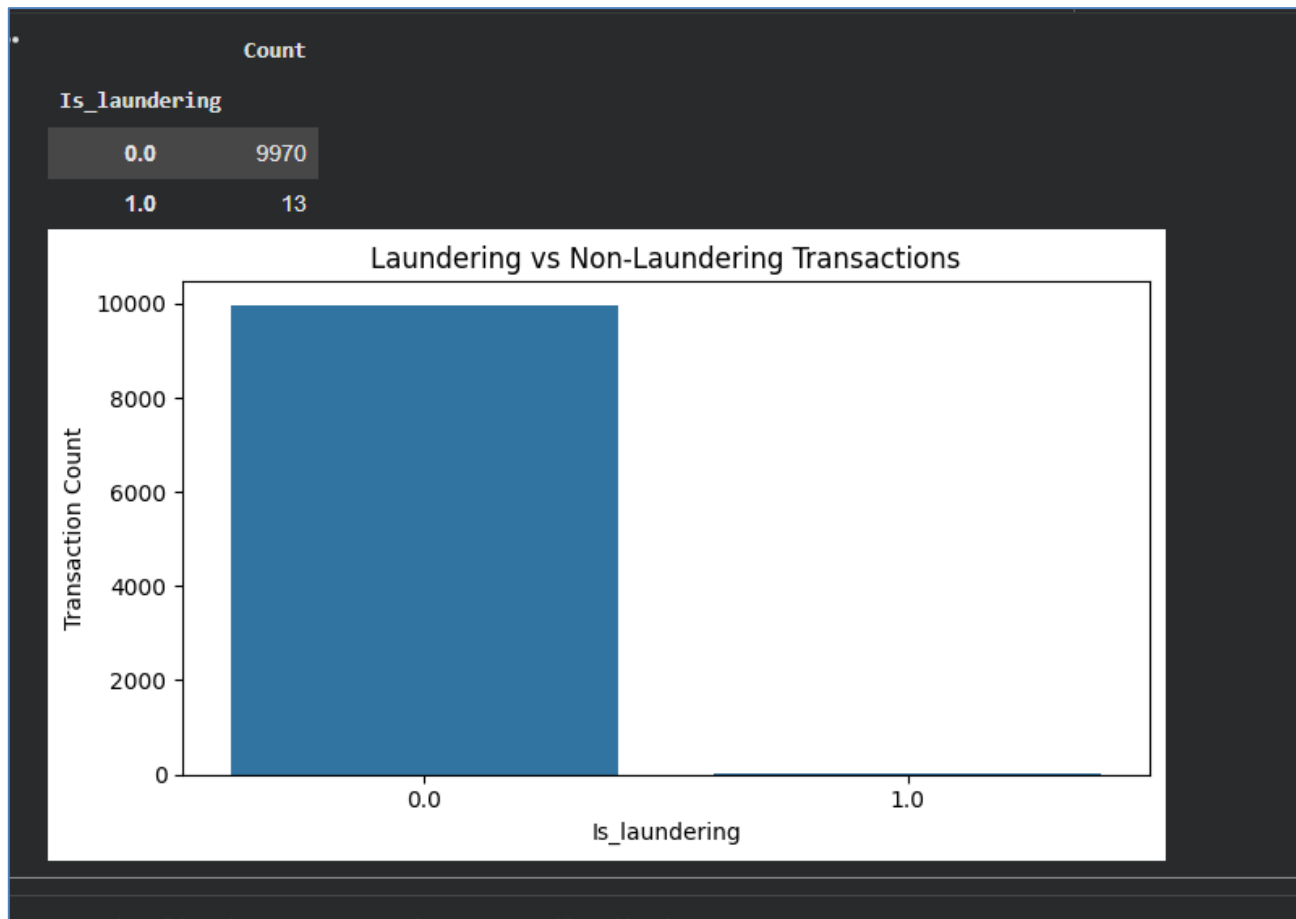
The dataset contains 13 unique payment currencies and 13 unique received currencies. Sender bank locations contain 17 unique values and receiver bank locations contain 18 unique values.

Data Visualization:-

The notebook includes visualizations for target distribution, transaction amount distribution, payment type, payment currency, laundering classification by payment type, a correlation heatmap and an interactive Plotly chart.

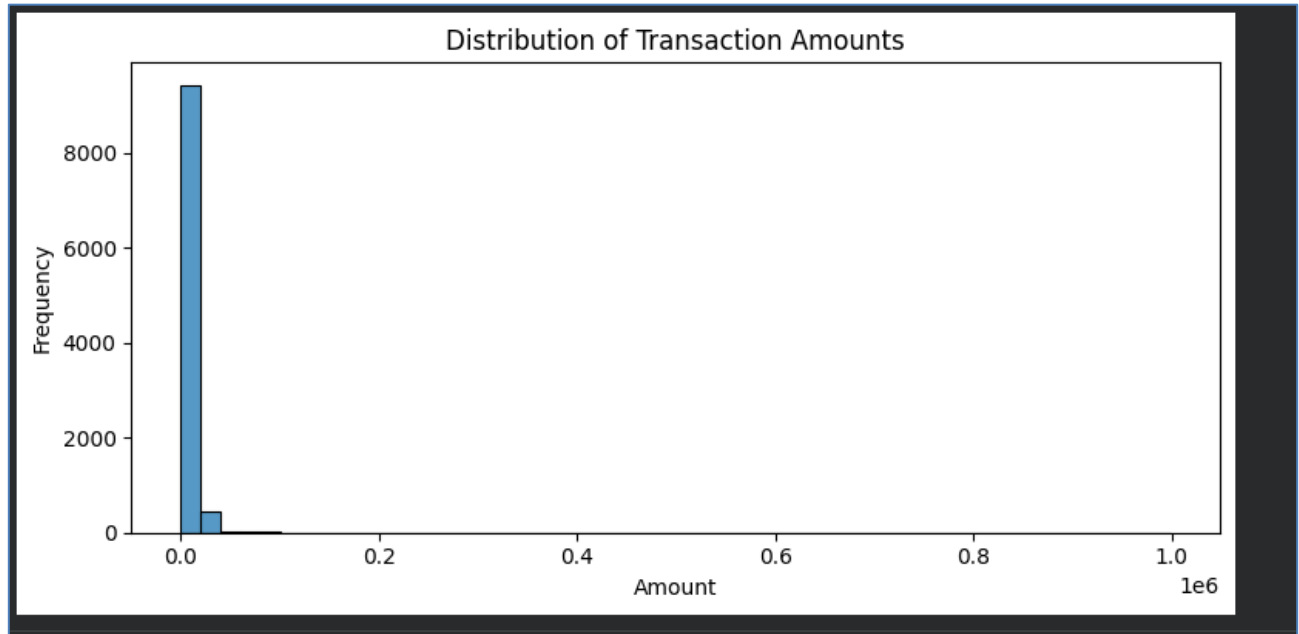
Laundering vs Non-Laundering

The target distribution visualization highlights the strong class imbalance: 9,970 non-laundering records versus 13 laundering records.



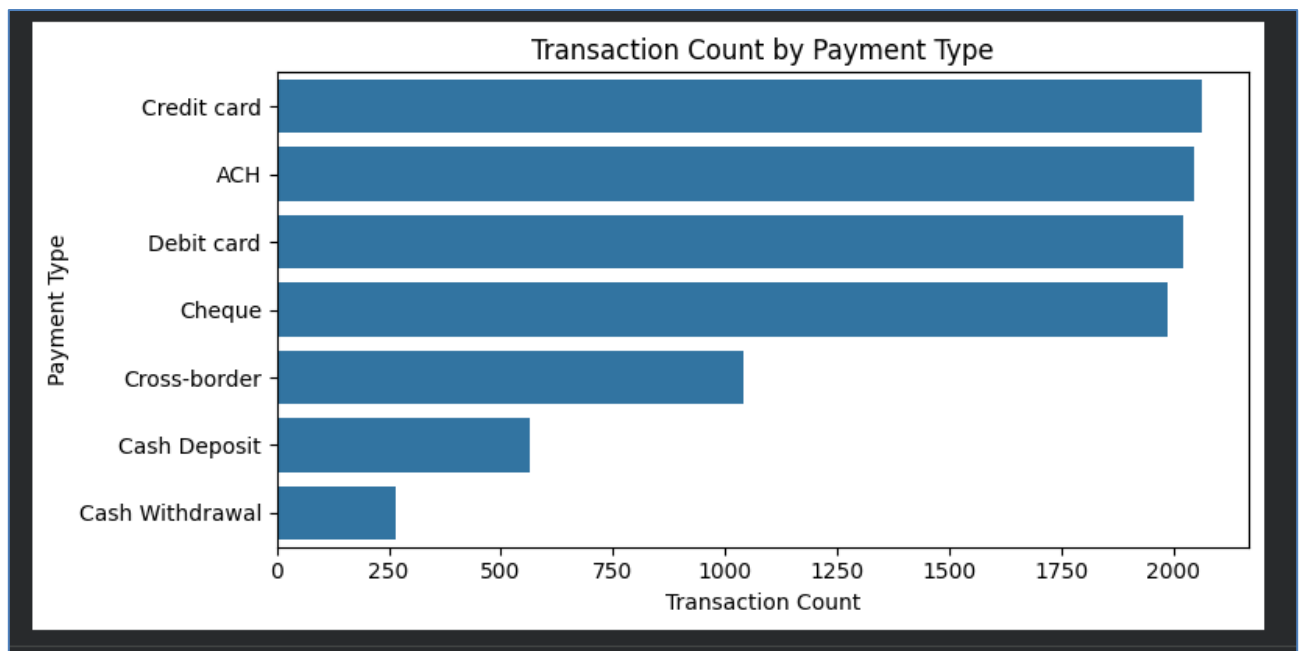
Distribution of Transaction Amounts

The histogram shows the distribution of transaction amounts and the presence of high-value observations.



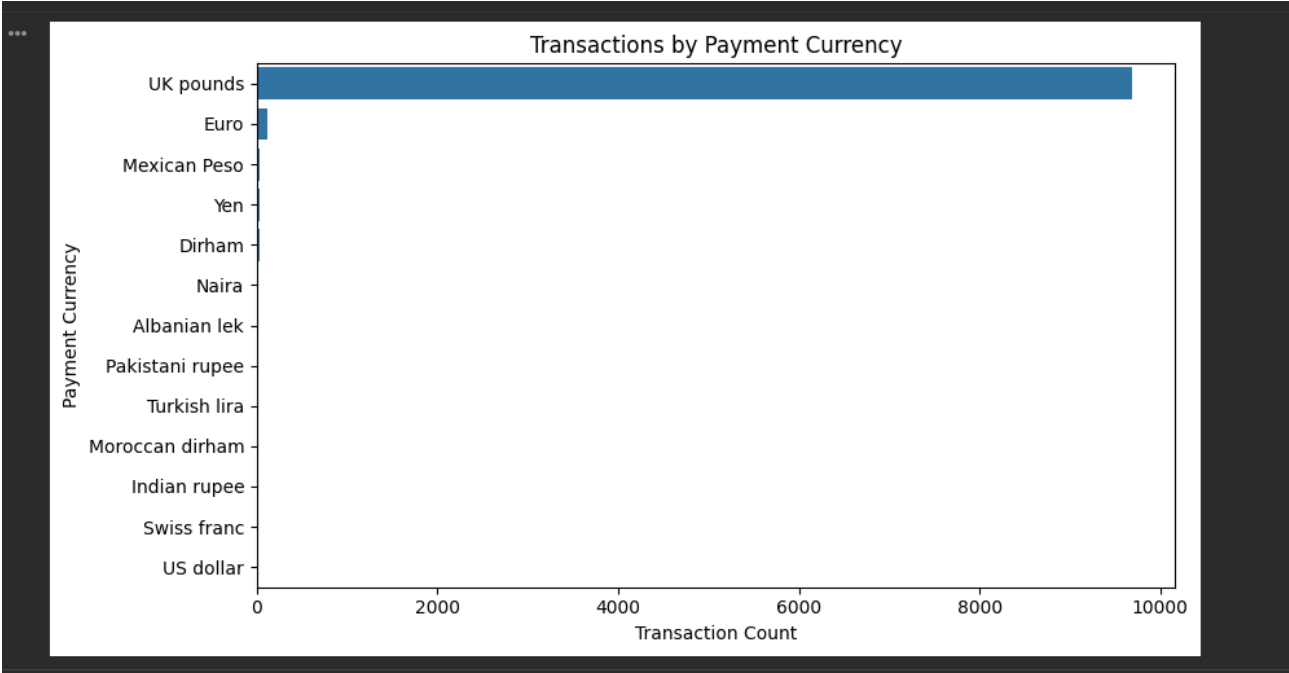
Transaction Count by Payment Type

The payment-type chart compares transaction volumes across the seven payment categories.



Transactions by Payment Currency

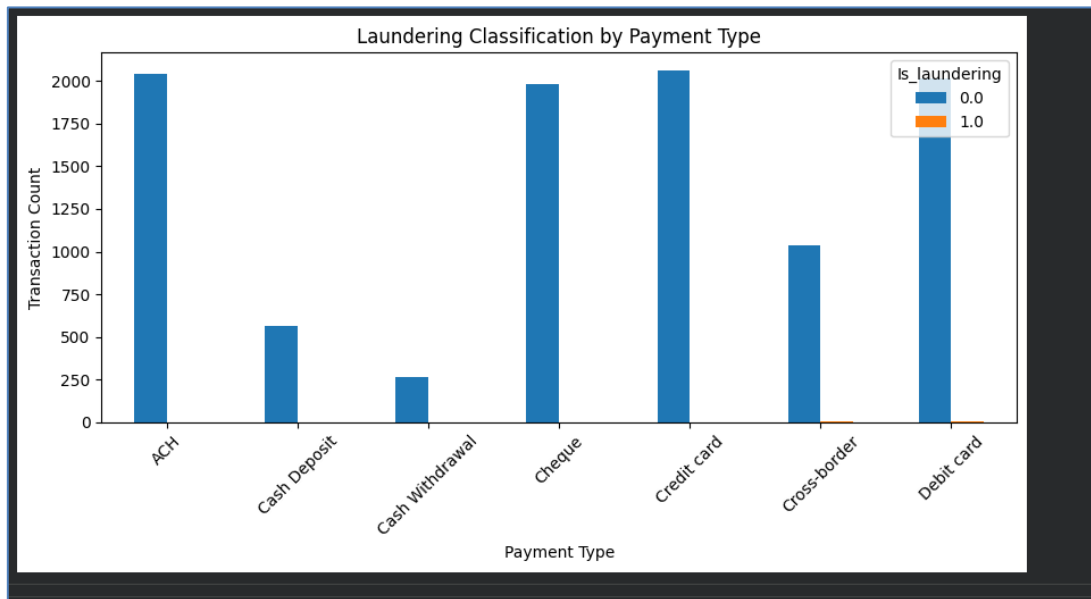
The currency visualization compares transaction counts across payment currencies.



Additional Visualizations:-

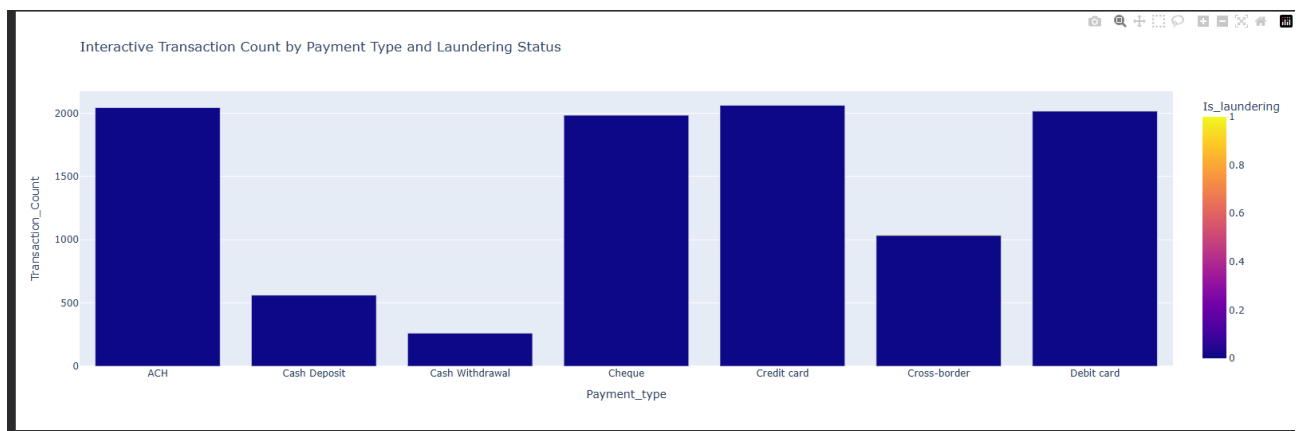
Laundering Classification by Payment Type

- This chart compares laundering classification across payment types.



Interactive Plotly Visualization

- The interactive Plotly chart displays transaction counts by payment type and laundering status and allows the user to inspect categories interactively.



These visualizations support the EDA by presenting both categorical and numerical patterns in a form that can be interpreted more easily than raw tables.

Machine Learning Model

Problem Type:-

The project is treated as a binary classification problem because `Is_laundering` contains two classes: 0 for non-laundering and 1 for laundering.

Algorithm – Random Forest Classifier:-

A Random Forest Classifier was selected for the classification task. The model combines multiple decision trees and can capture non-linear relationships between transaction features and the target.

Input Features:-

The executed notebook uses `Amount`, `Payment_currency`, `Received_currency`, `Sender_bank_location`, `Receiver_bank_location`, `Payment_type`, `Amount_Risk` and `Payment_Risk` as input features.

Training and Testing:-

The data was divided into training and testing sets using an 80:20 split with `random_state=42` and stratification on the target.

The training portion was used to fit a Random Forest with 150 trees, balanced class weights and parallel processing.

Model Evaluation:-

The model was evaluated on 1,997 test records. The test set contained 1,994 non-laundering records and 3 laundering records.

Metric	Observed Result
Accuracy	0.997496 (99.75%)
Precision	0.0000
Recall	0.0000
F1-Score	0.0000

Confusion Matrix:-

The classification report shows that all 1,994 non-laundering test records were classified correctly, while all 3 laundering test records were missed. Therefore, the confusion matrix is:

	Predicted 0	Predicted 1
Actual 0	1994	0
Actual 1	3	0

Although the accuracy is high, precision, recall and F1-score for the laundering class are zero. This is important because the objective of AML detection is to identify the rare positive class, not only to classify the dominant class.

Results and Finding:-

1. The analyzed dataset contains 9,983 usable records after one record with a missing target value was removed.
2. The target is highly imbalanced: 9,970 records are non-laundering and only 13 records are laundering, corresponding to approximately 99.87% and 0.13% respectively.
3. Transaction amount has a mean of approximately 8,360.91, a median of approximately 5,821.66 and a maximum of 998,453.63, showing a wide range of transaction values.
4. Credit card is the most frequent payment type in the analyzed records, with 2,064 observations.
5. The Random Forest achieved 99.75% accuracy, but its precision, recall and F1-score for the laundering class were all 0.00 because the three laundering transactions in the test set were not identified.

Interpretation:-

The evaluation demonstrates why accuracy alone is not sufficient for a highly imbalanced AML classification problem. The model correctly classifies the dominant non-laundering class, but it fails to identify the rare laundering class in the test set. Future work should therefore focus on class imbalance handling, feature engineering, threshold selection and comparison of alternative models.

Conclusion:-

This project demonstrates a complete Data Science workflow for an Anti-Money Laundering transaction monitoring problem. The workflow covers dataset loading, data quality checks, preprocessing, feature engineering, exploratory data analysis, visualization, machine learning classification and model evaluation.

The analysis identified a strong class imbalance in the available transaction sample. A Random Forest Classifier was trained using transaction and engineered risk features. The model achieved high overall accuracy, but the zero recall for the laundering class shows that the current model configuration is not sufficient for reliably detecting rare suspicious transactions.

The project therefore demonstrates both the usefulness of machine learning for transaction analysis and the importance of selecting evaluation metrics that reflect the actual business objective.

Future Scope:-

- Use a larger and more representative sample containing more positive laundering examples.
- Apply class imbalance techniques such as resampling or suitable class-weight strategies.
- Compare Random Forest with Logistic Regression, Decision Tree and other classification algorithms.
- Perform hyperparameter tuning and threshold optimization.
- Develop probability-based risk scores for transaction monitoring.
- Add model explainability to understand why a transaction is flagged.
- Integrate the model with real-time transaction monitoring.
- Deploy the improved model as an enterprise AML monitoring solution.

References:-

1. Oztas, B. et al., Synthetic Anti-Money Laundering Dataset (SAML-D).
2. Kaggle – Synthetic Transaction Monitoring Dataset (SAML-D):
<https://www.kaggle.com/datasets/berkanoztas/synthetic-transaction-monitoring-dataset-aml>
3. Scikit-learn documentation: <https://scikit-learn.org/>
4. Pandas documentation: <https://pandas.pydata.org/>
5. Matplotlib documentation: <https://matplotlib.org/>
6. Seaborn documentation: <https://seaborn.pydata.org/>
7. Plotly documentation: <https://plotly.com/python/>